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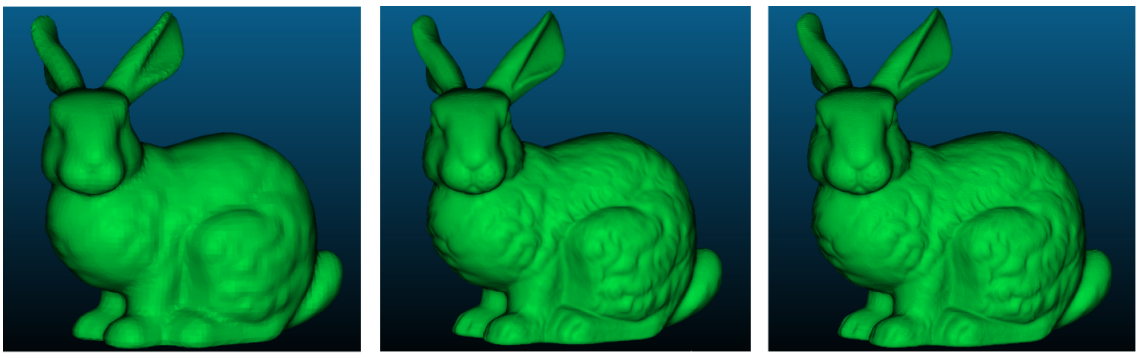
1 Question 1

I would say that a better reconstruction means that the object's structure and geometry is accurately preserved and reconstructed, and that the resulting surface is watertight and smooth. However, the surface should also contain fine details, and thus, it should neither be overly smoothed nor exhibit noisy artifacts. Our task is to find the reconstruction that is closest to this description.

The default parameters of PoissonRecon in CloudCompare are the following: (1) `octree depth` = 8, (2): `samples per node` = 1.5, (3) `point weight` = 2.0, and (4) `boundary` = Niemann. We will now plot and examine the effect of each parameter, and suggest the optimal ones.

First, we did not notice any visual difference when changing the `boundary` parameter across its three values (i.e., Free, Dirichlet, Niemann). Therefore, we keep the default one.

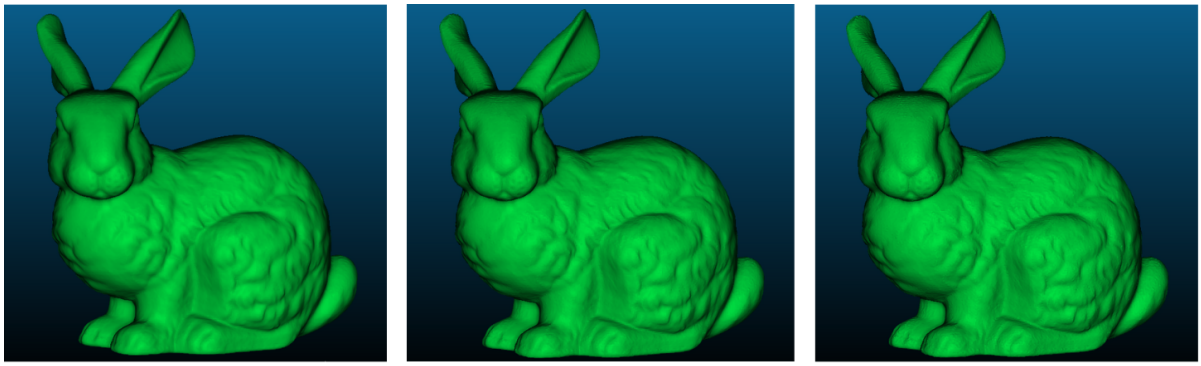
Second, we examine the `octree depth` parameter, whose effect is shown in Fig. 1. In general, this parameter controls the level of detail of the reconstruction as well as the smoothness of its surface. We observe that smaller values of the parameter lead to a loss of both shape information and surface detail. If we were to decrease it even more (e.g., to a value of 4), it would oversimplify the reconstruction and change the geometry of the object. In contrast, higher values increase surface detail, which, in our case, makes noise more apparent and leads to a grainier surface. Based on this, we choose to keep the default value of `octree depth = 8`.



(a) Octree depth = 6 (b) Octree depth = 8 (c) Octree depth = 12

Figure 1: Effect of the `octree depth` parameter on the surface reconstruction using PoissonRecon in CloudCompare. All other parameters are kept at their default values.

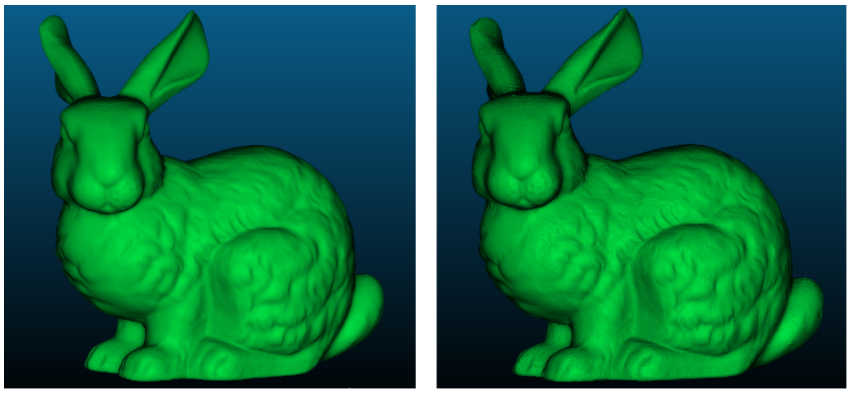
Next, we examine the `samples per node` parameter, whose effect is shown in Fig. 2. This parameter determines how many points fall and are averaged within each octree cell. We first observe that slightly increasing its value from the default one does not produce any noticeable visual difference. When we increase the value further (i.e., to 5), small granular artifacts begin to appear on the surface. According to the official documentation, higher values should be used for noisy input clouds. Since our point cloud is dense and exhibits no (or little) noise, increasing this parameter is unnecessary. Therefore, we keep the default value of `samples per node = 1.5`.



(a) samples per node = 1.5 (b) samples per node = 3 (c) samples per node = 5

Figure 2: Effect of the `samples per node` parameter on the surface reconstruction using PoissonRecon in CloudCompare. All other parameters are kept at their default values.

Lastly, we examine the `point weight` parameter, whose effect is shown in Fig. 3. Again, small increases in value did not yield noticeable changes, so we only plot and consider bigger increases. However, these values (i.e., a value of 8) result in an (even) more granular and less smooth surface. We therefore discard higher values and keep the default one of `point weight = 2.0`.



(a) point weight = 2.0 (b) point weight = 8.0

Figure 3: Effect of the point weight parameter on the surface reconstruction using PoissonRecon in CloudCompare. All other parameters are kept at their default values.

Finally, based on our previous observations, we decide that the default configuration of parameters, whose result is shown in Fig. 1 (b), provides the best reconstruction. This final mesh has 373998 triangles.

2 Question 2

Fig. 4 shows the surface reconstruction with the Hoppe function on a 128x128x128 voxel grid.

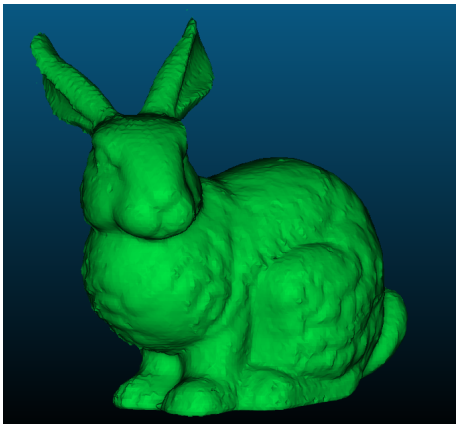


Figure 4: Surface reconstruction with the Hoppe function on a 128x128x128 voxel grid.

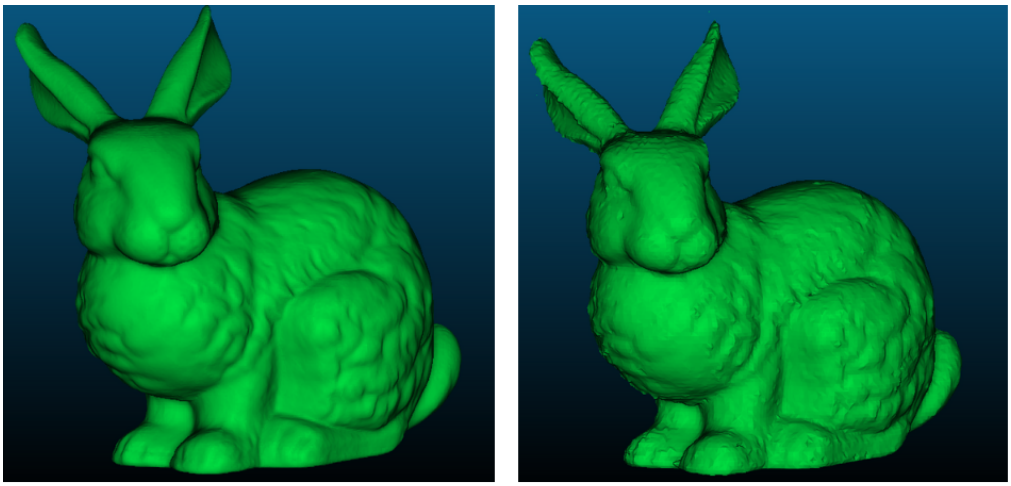
3 Question 3

In this question, we compare surface reconstructions obtained with PoissonRecon in CloudCompare and the Hoppe function. Table 3 presents a quantitative comparison between the two methods. The Hoppe method is approximately 20 times slower, which comes as a consequence of the 128^3 grid points requirig nearest-neighbor queries. Moreover, the Hoppe method produces around 4.5 times fewer triangles than the Poisson method.

Fig. 5 shows a qualitative comparison of two methods. We clearly observe artifacts in the result obtained with Hoppe function. The reconstructed surface appears grainy and significantly less smooth than the Poisson reconstruction, with noticeable pyramid-like structures and granular artifacts. Furthermore, due to the discontinuous nature of the Hoppe method, the reconstruction exhibits disconnected and unstable regions, particularly near the ears of the bunny (Fig. 6).

Reconstruction method	**Computation time (s)**	**Number of triangles**
Poisson (CloudCompare)	$\sim$2	373998
Hoppe function	40.8	82168

Table 1: Quantitative comparison between Poisson (CloudCompare) and Hoppe surface reconstruction.



(a) CloudCompare's Poisson reconstruction (b) Hoppe function reconstruction

Figure 5: Comparison between the reconstruction obtained with (a) PoissonRecon in CloudCompare and (b) Hoppe function.

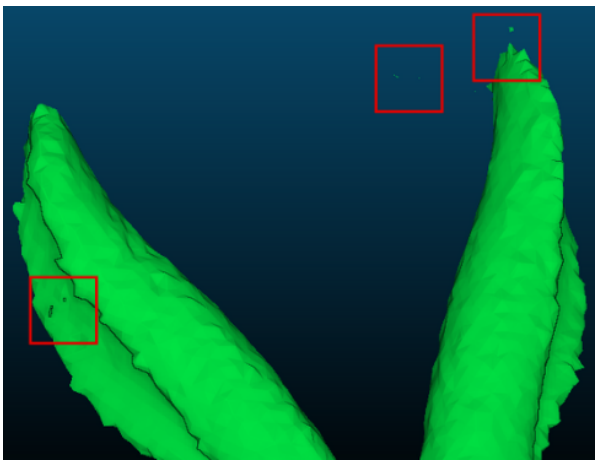


Figure 6: Discontinuous regions in the reconstruction obtained with Hoppe function.

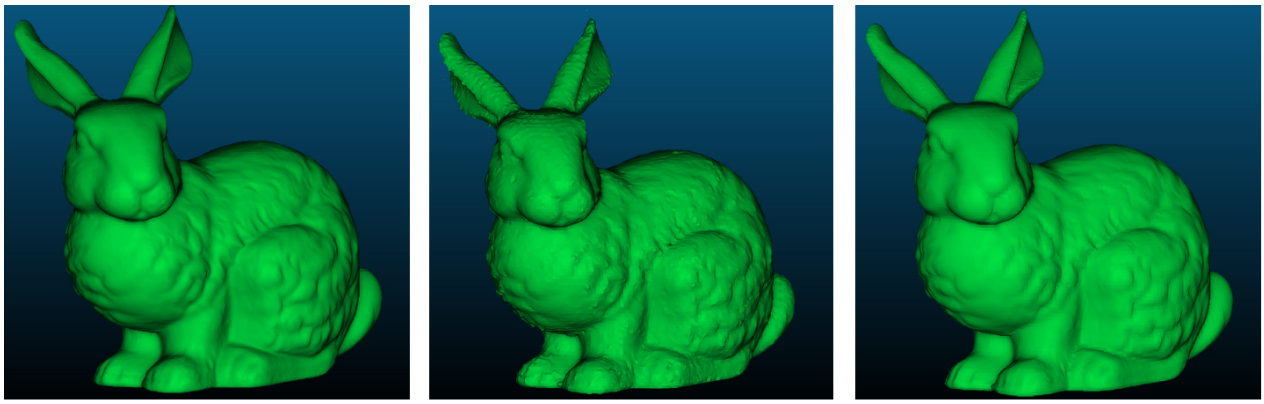
4 Question 4

In this question, we compare surface reconstructions obtained with PoissonRecon in CloudCompare, the Hoppe function, and the IMLS function. Table 4 presents a quantitative comparison between the three methods. Clearly, the IMLS method is the slowest, with an execution time approximately 1.75 times slower than the Hoppe method. This is because IMLS queries 30 neighboring points (instead of only one as in Hoppe) and performs additional computations (i.e., weighted sum and exponentiation). Moreover, this method yields the lowest number of triangles among all three.

Reconstruction method	**Computation time (s)**	**Number of triangles**
Poisson (CloudCompare)	$\sim$2	373998
Hoppe function	40.8	82168
IMLS function	70.3	78047

Table 2: Quantitative comparison between Poisson (CloudCompare), Hoppe and IMLS function surface reconstruction.

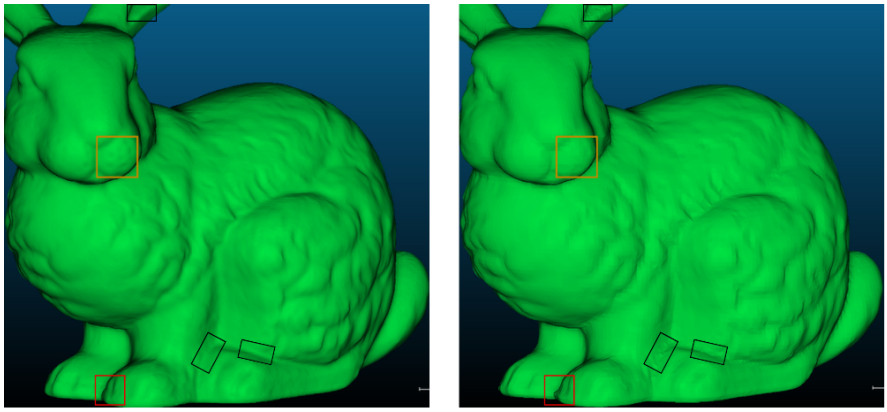
Fig. 7 shows a qualitative comparison of three methods. We observe that the IMLS reconstruction produces a smooth, non-grainy surface with significantly fewer artifacts compared to the Hoppe method. Moreover, since the IMLS implicit function is continuous, the reconstruction does not exhibit the disconnected regions we observed with Hoppe. Overall, the IMLS reconstruction is much closer to the Poisson reconstruction than Hoppe's one.



(a) CloudCompare's Poisson reconstruction (b) Hoppe function reconstruction (c) IMLS function reconstruction

Figure 7: Comparison between the reconstruction obtained with (a) PoissonRecon in CloudCompare, (b) Hoppe function and (c) IMLS function.

However, some local differences still remain, as shown Fig. 8. First, around the bunny's snout (nose), circular details are lost in the IMLS reconstruction (orange boxes). Second, in regions where the surface orientation changes sharply (i.e., between the foot and the leg, or between the inner and outer parts of the ear), the Poisson reconstruction is smoother, while the IMLS result shows slightly sharper transitions and small artifacts (black boxes). Lastly, the toe region shows a small tip/spike artifact (red box) in the IMLS reconstruction that is not present in the Poisson result and original point cloud.



(a) CloudCompare's Poisson reconstruction (b) IMLS function reconstruction

Figure 8: Local differences between Poison (CloudCompare) and IMLS function reconstruction.

In summary, we believe that for an object such as the bunny, which is globally smooth and regular, a global reconstruction method like Poisson produces more stable and visually consistent results than local methods such as Hoppe and IMLS, particularly in regions with strong changes in surface orientation.

5 Question 5

To obtain the reconstruction using Points2Surf, we first process and adapt our dataset (i.e. the Bunny point cloud) into the required format, which included normalization and text file creation, and then run the provided *full_eval.py* script. We use the *Max* model provided on the GitHub page, as recommended in the official implementation due to its smaller size and better accuracy. As for the parameters, we set `query_grid_resolution=128`, the neighborhood size to `epsilon=3`, the weighting parameter to `sigma=5`, and `certainty_threshold=0.5`. We run the code on Google Colab using an NVIDIA T4 GPU.

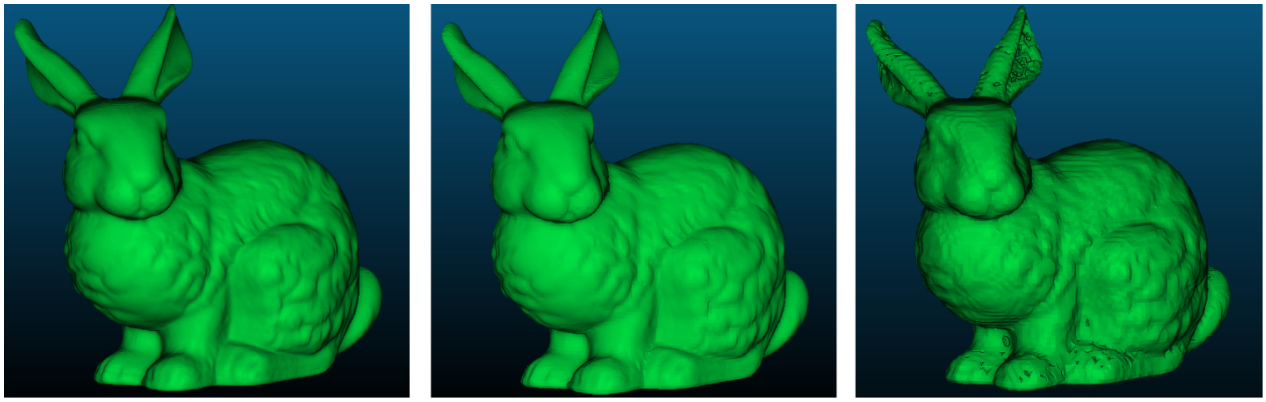
We now compare surface reconstructions obtained with PoissonRecon in CloudCompare, the IMLS function, and the Points2Surf method. Table 5 presents a quantitative comparison between the three approaches. As expected, Points2Surf is by far the slowest method, mainly due to the full neural network forward pass required to evaluate the signed distance function. This part of the inference took approximately 93 seconds, followed by sign propagation taking 19 second and marching cubes extraction taking 0.07s. Moreover, Points2Surf produces the lowest number of triangles among all previous methods.

Reconstruction method	**Computation time (s)**	**Number of triangles**
Poisson (CloudCompare)	~ 2	373998
IMLS function	70.3	78047
Points2Surf	~ 112	63114

Table 3: Quantitative comparison between Poisson (CloudCompare), IMLS function and Points2Surf surface reconstruction.

Fig. 9 shows a qualitative comparison of three methods. Unlike in previous cases, Points2Surf result in reconstruction with visible holes, particularly in the ear and leg regions. Moreover, its surface is not smooth and contains contour-like regions and sharp changes. Compared to the IMLS result, this reconstruction fails to preserve circular details on the bunny's snout, but, as a slight advantage, does not produce an artifact at the tip of the toe.

As a final note, I want to emphasize that better results could possibly be obtained with a different choice of parameters and/or model. Nevertheless, the bunny point cloud we used in this assignment represents a relatively regular and smooth shape, unlike the more irregular and incomplete examples shown in the Points2Surf paper and repository. While the authors of the paper note that Points2Surf leverages a learned prior on low-level patch shapes, which therefore improves the reconstruction accuracy in challenging scenarios, such advantages are less pronounced for our data. To conclude, based on our experiments, the Poisson and IMLS function reconstruction methods yield the best qualitative results, at least in the case of bunny point cloud.



(a) CloudCompare's Poisson reconstruction (b) IMLS function reconstruction c) Points2Surf reconstruction

Figure 9: Comparison between the reconstruction obtained with (a) PoissonRecon in CloudCompare and (b) IMLS function and (c) Points2Surf.